

Mode 8 — Square-BFP (centered / opt)

DP16 (16×16) → DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered add-and-square

Step 1 — DP16 (16×16)

$$\text{DP16 (16} \times \text{16)} = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP8 (16×16)

$$\text{DP16 (16} \times \text{16)} = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 (16} \times \text{16)}} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 (16} \times \text{16)}}$$

Step 3 — DP8 (16×8)

$$\text{DP16 (16} \times \text{16)} = 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 (16} \times \text{8)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8 (16} \times \text{8)}}$$

Step 4 — DP8 (16×4)

$$\begin{aligned} \text{DP16 (16} \times \text{16)} &= 2^{12} \underbrace{\sum_{i=0}^7 a_i b_i^{hh}}_{\text{DP8 (16} \times \text{4)}} + 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hl}}_{\text{DP8 (16} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{lh}}_{\text{DP8 (16} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{ll}}_{\text{DP8 (16} \times \text{4)}} \\ &+ 2^{12} \underbrace{\sum_{i=8}^{15} a_i b_i^{hh}}_{\text{DP8 (16} \times \text{4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i b_i^{hl}}_{\text{DP8 (16} \times \text{4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i b_i^{lh}}_{\text{DP8 (16} \times \text{4)}} + \underbrace{\sum_{i=8}^{15} a_i b_i^{ll}}_{\text{DP8 (16} \times \text{4)}} \end{aligned}$$

Step 5 — DP8 (8×4)

$$\begin{aligned} \text{DP16 (16} \times \text{16)} &= 2^{20} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{20} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{16} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + 2^{16} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} \\ &+ 2^{12} \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} \\ &+ 2^8 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} \\ &+ 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} \end{aligned}$$

Step 6 — Block exponents (one exponent block per DP8 (8×4))

$$\begin{aligned} M &= 2^{-\delta_0} \left(2^{20} \sum_{i=0}^7 a_i^{hi} b_i^{hh} + 2^{16} \sum_{i=0}^7 a_i^{hi} b_i^{hl} + 2^{12} \sum_{i=0}^7 a_i^{lo} b_i^{hh} + 2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{lh} \right. \\ &\quad \left. + 2^8 \sum_{i=0}^7 a_i^{lo} b_i^{hl} + 2^8 \sum_{i=0}^7 a_i^{hi} b_i^{ll} + 2^4 \sum_{i=0}^7 a_i^{lo} b_i^{lh} + \sum_{i=0}^7 a_i^{lo} b_i^{ll} \right) + 2^{-\delta_1} \left(\cdots i \in [8, 15] \cdots \right) \end{aligned}$$

$E_k = E_{a,k} + E_{b,k}$; $E = \max(E_0, E_1)$, $\delta_k = E - E_k$; truncating shifts, on block sums only; DP16 (16×16) = $M \cdot 2^E$.
Step 7 — DP8 (4×4)

$$\begin{aligned}
M = & 2^{-\delta_0} 2^{24} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_0} 2^{20} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{24} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{20} \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{hh}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{20} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_0} 2^{16} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{20} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{16} \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{16} \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_0} 2^{12} \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{16} \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{12} \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{16} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_0} 2^{12} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{16} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{12} \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{12} \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_0} 2^8 \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{12} \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^8 \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{12} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_0} 2^8 \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^{12} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^8 \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_0} 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^8 \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^4 \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_0} \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^{-\delta_1} \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

Step 8 — Centered add-and-square (subtract 8 from unsigned nibbles; per DP8 (4×4) add $c = 0/512/2048$ for $0/1/2$ unsigned nibbles)

$$\begin{aligned}
M = & 2^{-\delta_0} 2^{24} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + b_i^{hh})^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{24} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hh})^2 - \sum_{i=8}^{15} (a_{H,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{hh})^2 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + b_i^{hh})^2 - \sum_{i=8}^{15} (a_{L,i}^{hi} - 8)^2 - \sum_{i=8}^{15} (b_i^{hh} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

$$\begin{aligned}
& + 2^{-\delta_0} 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{hl} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{hl} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{hl} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 - \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{hi} + (b_i^{hl} - 8))^2 - \sum_{i=8}^{15} (a_{H,i}^{hi} - 8)^2 - \sum_{i=8}^{15} (b_i^{hl} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + (b_i^{hl} - 8))^2 - \sum_{i=8}^{15} (a_{L,i}^{hi} - 16)^2 - \sum_{i=8}^{15} (b_i^{hl} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + b_i^{hh})^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 8)^2 - \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + b_i^{hh})^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 8)^2 - \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + b_i^{hh})^2 - \sum_{i=8}^{15} (a_{H,i}^{lo} - 8)^2 - \sum_{i=8}^{15} (b_i^{hh} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + b_i^{hh})^2 - \sum_{i=8}^{15} (a_{L,i}^{lo} - 8)^2 - \sum_{i=8}^{15} (b_i^{hh} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{lh} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{lh} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{lh} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 - \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{hi} + (b_i^{lh} - 8))^2 - \sum_{i=8}^{15} (a_{H,i}^{hi} - 8)^2 - \sum_{i=8}^{15} (b_i^{lh} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + (b_i^{lh} - 8))^2 - \sum_{i=8}^{15} (a_{L,i}^{hi} - 16)^2 - \sum_{i=8}^{15} (b_i^{lh} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{hl} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{hl} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + (b_i^{hl} - 8))^2 - \sum_{i=8}^{15} (a_{H,i}^{lo} - 16)^2 - \sum_{i=8}^{15} (b_i^{hl} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

$$\begin{aligned}
& + 2^{-\delta_1} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + (b_i^{hl} - 8))^2 - \sum_{i=8}^{15} (a_{L,i}^{lo} - 16)^2 - \sum_{i=8}^{15} (b_i^{hl} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{ll} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{ll} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{ll} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 - \sum_{i=0}^7 (b_i^{ll} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{hi} + (b_i^{ll} - 8))^2 - \sum_{i=8}^{15} (a_{H,i}^{hi} - 8)^2 - \sum_{i=8}^{15} (b_i^{ll} - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + (b_i^{ll} - 8))^2 - \sum_{i=8}^{15} (a_{L,i}^{hi} - 16)^2 - \sum_{i=8}^{15} (b_i^{ll} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{lh} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{lh} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + (b_i^{lh} - 8))^2 - \sum_{i=8}^{15} (a_{H,i}^{lo} - 16)^2 - \sum_{i=8}^{15} (b_i^{lh} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + (b_i^{lh} - 8))^2 - \sum_{i=8}^{15} (a_{L,i}^{lo} - 16)^2 - \sum_{i=8}^{15} (b_i^{lh} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{ll} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{ll} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_0} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{ll} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{ll} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + (b_i^{ll} - 8))^2 - \sum_{i=8}^{15} (a_{H,i}^{lo} - 16)^2 - \sum_{i=8}^{15} (b_i^{ll} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{-\delta_1} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + (b_i^{ll} - 8))^2 - \sum_{i=8}^{15} (a_{L,i}^{lo} - 16)^2 - \sum_{i=8}^{15} (b_i^{ll} - 16)^2 + 2048 \right)}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

Step 9 — All components

$$\begin{aligned}
\text{PE} = & 2^{-\delta_0} 2^{24} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hh})^2 + 2^{-\delta_0} 2^{20} \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + b_i^{hh})^2 + 2^{-\delta_1} 2^{24} \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hh})^2 + 2^{-\delta_1} 2^{20} \sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + b_i^{hh})^2 \\
& + 2^{-\delta_0} 2^{20} \sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{hl} - 8))^2 + 2^{-\delta_0} 2^{16} \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{hl} - 8))^2 + 2^{-\delta_1} 2^{20} \sum_{i=8}^{15} (a_{H,i}^{hi} + (b_i^{hl} - 8))^2 + 2^{-\delta_1} 2^{16} \sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + (b_i^{hl} - 8))^2 \\
& + 2^{-\delta_0} 2^{16} \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + b_i^{hh})^2 + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + b_i^{hh})^2 + 2^{-\delta_1} 2^{16} \sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + b_i^{hh})^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + b_i^{hh})^2 \\
& + 2^{-\delta_0} 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{lh} - 8))^2 + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{lh} - 8))^2 + 2^{-\delta_1} 2^{16} \sum_{i=8}^{15} (a_{H,i}^{hi} + (b_i^{lh} - 8))^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + (b_i^{lh} - 8))^2 \\
& + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{hl} - 8))^2 + 2^{-\delta_0} 2^8 \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{hl} - 8))^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + (b_i^{hl} - 8))^2 + 2^{-\delta_1} 2^8 \sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + (b_i^{hl} - 8))^2 \\
& + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{ll} - 8))^2 + 2^{-\delta_0} 2^8 \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{ll} - 8))^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} (a_{H,i}^{hi} + (b_i^{ll} - 8))^2 + 2^{-\delta_1} 2^8 \sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + (b_i^{ll} - 8))^2 \\
& + 2^{-\delta_0} 2^8 \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{lh} - 8))^2 + 2^{-\delta_0} 2^4 \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{lh} - 8))^2 + 2^{-\delta_1} 2^8 \sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + (b_i^{lh} - 8))^2 + 2^{-\delta_1} 2^4 \sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + (b_i^{lh} - 8))^2 \\
& + 2^{-\delta_0} 2^4 \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{ll} - 8))^2 + 2^{-\delta_0} \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{ll} - 8))^2 + 2^{-\delta_1} 2^4 \sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + (b_i^{ll} - 8))^2 + 2^{-\delta_1} \sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + (b_i^{ll} - 8))^2 \\
\alpha = & 2^{-\delta_0} 2^{24} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{-\delta_0} 2^{20} \sum_{i=0}^7 (a_{L,i}^{hi} - 8)^2 + 2^{-\delta_1} 2^{24} \sum_{i=8}^{15} (a_{H,i}^{hi})^2 + 2^{-\delta_1} 2^{20} \sum_{i=8}^{15} (a_{L,i}^{hi} - 8)^2 \\
& + 2^{-\delta_0} 2^{20} \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 + 2^{-\delta_0} 2^{16} \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 + 2^{-\delta_1} 2^{20} \sum_{i=8}^{15} (a_{H,i}^{hi} - 8)^2 + 2^{-\delta_1} 2^{16} \sum_{i=8}^{15} (a_{L,i}^{hi} - 16)^2 \\
& + 2^{-\delta_0} 2^{16} \sum_{i=0}^7 (a_{H,i}^{lo} - 8)^2 + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 (a_{L,i}^{lo} - 8)^2 + 2^{-\delta_1} 2^{16} \sum_{i=8}^{15} (a_{H,i}^{lo} - 8)^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} (a_{L,i}^{lo} - 8)^2 \\
& + 2^{-\delta_0} 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 + 2^{-\delta_1} 2^{16} \sum_{i=8}^{15} (a_{H,i}^{hi} - 8)^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} (a_{L,i}^{hi} - 16)^2 \\
& + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 + 2^{-\delta_0} 2^8 \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} (a_{H,i}^{lo} - 16)^2 + 2^{-\delta_1} 2^8 \sum_{i=8}^{15} (a_{L,i}^{lo} - 16)^2 \\
& + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 + 2^{-\delta_0} 2^8 \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} (a_{H,i}^{hi} - 8)^2 + 2^{-\delta_1} 2^8 \sum_{i=8}^{15} (a_{L,i}^{hi} - 16)^2 \\
& + 2^{-\delta_0} 2^8 \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 + 2^{-\delta_0} 2^4 \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 + 2^{-\delta_1} 2^8 \sum_{i=8}^{15} (a_{H,i}^{lo} - 16)^2 + 2^{-\delta_1} 2^4 \sum_{i=8}^{15} (a_{L,i}^{lo} - 16)^2 \\
& + 2^{-\delta_0} 2^4 \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 + 2^{-\delta_0} \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 + 2^{-\delta_1} 2^4 \sum_{i=8}^{15} (a_{H,i}^{lo} - 16)^2 + 2^{-\delta_1} \sum_{i=8}^{15} (a_{L,i}^{lo} - 16)^2 \\
\beta = & 2^{-\delta_0} 2^{24} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{-\delta_0} 2^{20} \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 2^{-\delta_1} 2^{24} \sum_{i=8}^{15} (b_i^{hh})^2 + 2^{-\delta_1} 2^{20} \sum_{i=8}^{15} (b_i^{hh} - 8)^2 \\
& + 2^{-\delta_0} 2^{20} \sum_{i=0}^7 (b_i^{hl} - 8)^2 + 2^{-\delta_0} 2^{16} \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2^{-\delta_1} 2^{20} \sum_{i=8}^{15} (b_i^{hl} - 8)^2 + 2^{-\delta_1} 2^{16} \sum_{i=8}^{15} (b_i^{hl} - 16)^2 \\
& + 2^{-\delta_0} 2^{16} \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 2^{-\delta_1} 2^{16} \sum_{i=8}^{15} (b_i^{hh} - 8)^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} (b_i^{hh} - 8)^2 \\
& + 2^{-\delta_0} 2^{16} \sum_{i=0}^7 (b_i^{lh} - 8)^2 + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2^{-\delta_1} 2^{16} \sum_{i=8}^{15} (b_i^{lh} - 8)^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} (b_i^{lh} - 16)^2 \\
& + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2^{-\delta_0} 2^8 \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} (b_i^{hl} - 16)^2 + 2^{-\delta_1} 2^8 \sum_{i=8}^{15} (b_i^{hl} - 16)^2 \\
& + 2^{-\delta_0} 2^{12} \sum_{i=0}^7 (b_i^{ll} - 8)^2 + 2^{-\delta_0} 2^8 \sum_{i=0}^7 (b_i^{ll} - 16)^2 + 2^{-\delta_1} 2^{12} \sum_{i=8}^{15} (b_i^{ll} - 8)^2 + 2^{-\delta_1} 2^8 \sum_{i=8}^{15} (b_i^{ll} - 16)^2 \\
& + 2^{-\delta_0} 2^8 \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2^{-\delta_0} 2^4 \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2^{-\delta_1} 2^8 \sum_{i=8}^{15} (b_i^{lh} - 16)^2 + 2^{-\delta_1} 2^4 \sum_{i=8}^{15} (b_i^{lh} - 16)^2 \\
& + 2^{-\delta_0} 2^4 \sum_{i=0}^7 (b_i^{ll} - 16)^2 + 2^{-\delta_0} \sum_{i=0}^7 (b_i^{ll} - 16)^2 + 2^{-\delta_1} 2^4 \sum_{i=8}^{15} (b_i^{ll} - 16)^2 + 2^{-\delta_1} \sum_{i=8}^{15} (b_i^{ll} - 16)^2 \\
C = & (2^{-\delta_0} + 2^{-\delta_1}) (0 + 2^{29} + 2^{29} + 2^{27} + 2^{25} + 2^{21} + 2^{25} + 2^{23} + 2^{23} + 2^{19} + 2^{21} + 2^{19} + 2^{19} + 2^{15} + 2^{15} + 2^{11}) \\
& = (2^{-\delta_0} + 2^{-\delta_1}) \cdot 1297680384 \\
M = & \frac{1}{2} (\text{PE} - \alpha - \beta + C), \quad E = \max(E_0, E_1)
\end{aligned}$$